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Computer Codes and Arithmetic

Decimal system:

The decimal system uses a base of 10 and thus it uses 10 symbols, 0 to 9.

Example: $\overset{3\ 2\ 1\ 0}{5704} \leftarrow$ Position

$= 5\times10^3 + 7\times10^2 + 0\times10^1 + 4\times10^0$

$= 5000 + 700 + 0 + 4$

$= 5704$

Binary system:

The binary system is the positional number system to the base 2. It uses two symbols 0 and 1.

Example: $1101 = 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 8 + 4 + 0 + 1$

$= 13$

$1101_2 = 13_{10}$

Hexadecimal system:

The hexadecimal system is a number system that uses 16 as its base. It uses 16 symbols 0 to 9, A B, C, D, E, F

Example: $12AF = 1\times16^3 + 2\times16^2 + 10\times16^1 + 15\times16^0$

$= 4096 + 512 + 160 + 15$

$12AF_{16} = 4783_{10} = 4783$

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Octal System:

The octal number system is a system having base 8. The eight octal symbols are 0 to 7.

Example: $2056_8 = 2\times8^3 + 0\times8^2 + 5\times8^1 + 6\times8^0$

$= 1024 + 0 + 40 + 6$

$= 1070$

$2056_8 = 1070_{10}$

Conversion of number

1. non - decimal to decimal system
2. decimal to non - decimal system
3. octal to hexadecimal system
4. hexadecimal to octal system

1. Non - decimal to decimal system:

Binary to decimal : $(1101.1101)_2 = (?)_{10}$

⇒ Integral Part = $(1101)_2$

$= 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 13$

Fractional Part = $(.1101)_2$

$= 1\times2^{-1} + 1\times2^{-2} + 0\times2^{-3} + 1\times2^{-4}$

$= 0.8125$

$\therefore (1101.1101)_2 = (13.8125)_{10}$

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Hexadecimal to decimal: $(12AF)_{16} = (?)_{10}$

$A = 10$
$F = 15$

$12AF = 1\times16^3 + 2\times16^2 + 10\times16^1 + 15\times16^0$

$= (4783)_{10}$

2. Decimal to non-decimal system:

Decimal to binary: $(43.375)_{10} = (?)_2$

Integral Part:

2 | 43	
2 | 21	1
2 | 10	1
2 | 5	0
2 | 2	1
2 | 1	0
	0 1

Fractional Part:

	.375 × 2
0	.75 × 2
1	.5 × 2
1	.0

$(43.375)_{10} = (101011.011)_2$

Decimal to octal: $(163)_{10} = (?)_8$

8 | 163	
8 | 20	3
8 | 2	4
	0 2

$(163)_{10} = (243)_8$

Decimal to hexadecimal: $(157)_{10} = (?)_{16}$

16 | 157	
16 | 9	13
	0 9

$(157)_{10} = (9D)_{16}$

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3. Octal to Hexadecimal: $(243)_8 = (?)_{16}$

2 4 3

010 100 011

10 3

$2^3 = 8$

A = 10

$(243)_8 = (0A3)_{16}$

4. Hexadecimal to Octal: $(39.BB)_{16} = (?)_8$

3 9 . B B

0011 1001 . 1011 1000

0 7 1 5 6

$= (071.56)_4$

Representation of number.

largest number $= 2^n - 1$

where n is the word length in bits

Example:

101, bit n = 3

$\therefore$ largest number $= 2^3 - 1 =$

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Example :- Represent -13 in binary form.

⇒
2 | 13
2 | 6 - 1
2 | 3 - 0
2 | 1 - 1
0 - 1

$(13)_{10} = (1101)_2$

13 = 01101

1's complement = 1 0010

+ 1

2's complement = 10011

0 → +ve
1 → -ve

13 = -10011

sign bit

Example :- -32768

⇒ -32768 = -32767 + (-1)

32767 = 0111 1111 1111 1111

1's compⁿ = 1000 0000 0000 0000

+0000 0000 0000 0001

2's compⁿ = 1000 0000 0000 0001

1 = 0000 0000 0000 0001

1's compⁿ = 1111 1111 1111 1110

+0000 0000 0000 0001

2's compⁿ = 1111 1111 1111 1111

-32767 + (-1) ⇒ 1000 0000 0000 0001

1111 1111 1111 1111

-32768 = 1 1000 0000 0000 0000

Sign bit

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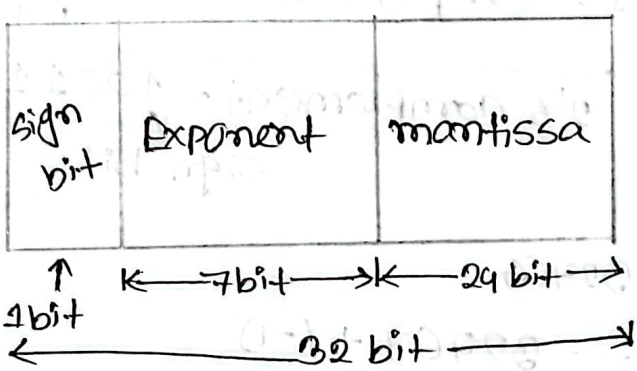
Floating Point Representation:

Floating point form representation is

$$n = f \times 10^E \qquad 0.1 \le |f| < 1$$

where f is called mantissa and E is the exponent.



Example:

0.00596 is expressed as $0.5956 \times 10^{-2} \rightarrow 0.5956E-2$

65.7452 is expressed as $0.657452 \times 10^{2} \rightarrow 0.657452E2$

-486.8 is expressed as -0.4868×10^{3}

Computer arithmetic:

Integer Arithmetic:

Floating point Arithmetic:

Example: 3.12: Add the numbers 0.064572 E2 and 0.586351 E5

⇒ Let $x = 0.586351$ E5

$y = 0.064572$ E2

$= \frac{0.064572 \times 1000}{1000}$ (E3) E2

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$= 0.0000964572\ E5$

$E_z = 5$

$f_z = f_x + f_y$

$= 0.586351 + 0.0000964572$

$= 0.5874315\ E5$

Example: 3.14 Add the numbers $0.735816\ E4$ and $0.635742\ E4$

$\Rightarrow E_z = 4$

$f(x) = 0.735816E4 \quad f_y = 0.7635742\ E4$

$f_z = 0.735816 + 0.635742$

$= 1.371558\ E4$

$= 0.1371558\ E5$

Example-3.15: subtract $0.0994576\ E-3$ from $0.999658\ E-3$

$\Rightarrow$ Let $x = 0.999658\ E-3$

$y = 0.0994576\ E-3$

$f_z = 0.999658 - 0.0994576$

$= 0.005082\ E-3$

$= 0.5082\ E-5$

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Add the numbers 0.735816 E4 and 0.635742 E-2

$\Rightarrow x = 0.735816\ E4$

$y = 0.635742\ E-2$

$10^6 \times 10^{-2} = 10^4 = E4$

$= \frac{0.635742 \times 1000000}{1000000}\ E-2$

$= 0.000000635742\ E4$

$Ez = 4$

$fz = fx + fy$

$= 0.735816 + 0.000000635742$

$= 0.735816635742\ E4$

Example 3.16- Multiply the numbers 0.200000 E4 and 0.400000 E2

$\Rightarrow fz = 0.200000 \times 0.400000$

$= 0.080000$

$Ez = 4 - 2 = 2$

$z = 0.080000\ E2 = 0.80000\ E1$

Example-3.17: Divide the number 0.876543 E-5 by 0.200000 E-3

$\Rightarrow fz = 0.876543 \div 0.200000$

$= 4.382715$

$Ez = -5 - (-3) = -2$

$z = 4.382715\ E-2 = 0.4382715\ E-1$

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Precision:

Example-4.1:

a) 4.3201 → .4320 1 $\times 10^{-4}$ ∴ Precision = 10^{-4}

b) 4.32 → 0.432×10^{2} ∴ Precision = 10^{-2}

c) 4.320106 → .4320106 ∴ Precision = 10^{-6}

∴ The last number has the greatest precision.

Significant number:

Example - 4.2:

a) 95.763 significant number 5

b) 0.008972 significant number 4

c) 0.0456000 " " 6

d) 36 " " 2

e) 3600 " " Not satisfied

f) 3600.00 " " 6

Inherent Error (Input error)

Inherent error are those that one are present in the data supplied to the model.

It contain 2 components:

1. Data errors.
2. Conversion errors.

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Data error :

Data error arises from experimental measurement and is unavoidable due to limitations of instruments and observation.

Experimental data → limited accuracy → unavoidable error.

(কোনো Problem এর data experiment / measurement থেকে নেয়া হলে এই error হয়)

Conversion error :

Conversion error arises because a computer cannot store all digits exactly, leading to round-off error.

Limited Storage → digit cut → round-off error.

Computer যখন সব digit store করতে পারে না তখন কিছু digit cut করে দেয়। এর ফলে Conversion/representation error হয়।

Example - 4.3 :

Represent the decimal numbers 0.1 and 0.4 in binary form with an accuracy of 8 binary digit. Add them and then convert the result back to the decimal form.

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$\Rightarrow (0.1)_{10} \rightarrow$

	.1
	x2
0	.2
	x2
0	.4
	x2
0	.8
	x2
1	.6
	x2
1	.2
	x2
0	.4
	x2
0	.8
	x2
1	.6

$(0.4)_{10} \rightarrow$

	.4
	x2
0	.8
	x2
1	.6
	x2
1	.2
	x2
0	.4
	x2
0	.8
	x2
1	.6
	x2
1	.2
	x2
0	.4

$0.1 = 0.00011001$

$0.4 = 0.01100110$

$0.5 = 0.01111111$

$(0.01111111)_2 = 0\times2^{-1} + 1\times2^{-2} + 1\times2^{-3} + 1\times2^{-4} + 1\times2^{-5} + 1\times2^{-6} + 1\times2^{-7} + 1\times2^{-8}$

$= (0.49609375)_{10} \neq 0.5$

∴ This is conversion error.

Truncation Error :

Truncation error arise from using an approximation in place of an exact mathematical procedure.

Error formula = (True value − Approximate value)

or,

error : Remaining terms after truncation.

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(এটা হল যখন আমরা exact mathematical Process অনুসরণ না করে, ছোট করে ব্যবহার করি, যেমন যেমন অঙ্ক থেকে লাগে হয়)

Example: 4.5 - Find the truncation error in the result of the following function for $x = \frac{1}{5}$ when we use a) first three tearm. b) first four tearm. c) first five tearm.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

⇒ (a) Approximation, $S_3 = 1 + x + \frac{x^2}{2!}$

Truncation error : True value - Approximation value

$$= \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

$$= \frac{(0.2)^3}{3!} + \frac{(0.2)^4}{4!} + \frac{(0.2)^5}{5!} + \frac{(0.2)^6}{6!}$$

$$= 0.1402755 \times 10^{-2}$$

(b) Approximation, $S_4 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$

Truncation error = T.V - A.V

$$= \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

$$= 0.604122 \times 10^{-4}$$

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① Approximate, $S_5 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$

T.E = T.V − A.V

$= \frac{x^5}{5!} + \frac{x^6}{6!}$

$= 0.2755 \times 10^{-5}$

Numerical Error:

Numerical errors are introduced during the process of implementation of a numerical method. They come from two form:

1. Round-off errors.
2. Truncation errors.

Round-off error:

occurs when a fixed number of digit are used to represent exact numbers.

Rounding a number can be done in two way.

① Chopping (extra digit মুছে)

② Symmetric rounding (নিকটতম value তে যাবে)

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Chopping :

True value $(f + g \times 10^{-d}) \times 10^{E}$

approximate value $f \times 10^{E}$

error $\rightarrow g \times 10^{(E-d)}$

$f \rightarrow$ mantissa

$g \rightarrow$ dropped part. (কাটা বাদ দেওয়া part)

$d \rightarrow$ digit length.

$E \rightarrow$ Exponent.

Symmetric round-off :

True value $= f \times 10^{E} + g \times 10^{(E-d)}$

Approximate value $= f \times 10^{E}$

error $\rightarrow g \times 10^{E-d}$

} যদি $g < 5$

Rule:

যদি, $g < 5$ হয়; তাহলে no change.

$\therefore 42.784_{g} \rightarrow 42.78$

যদি $g \geq 5$ হলে last digit + 1 হবে,

$\therefore 42.786_{g} \rightarrow 42.79$

তাহলে, Approx $\rightarrow f + 10^{-d} \times 10^{E}$

Error $\rightarrow (g-1) \times 10^{(E-d)}$

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Example-4.4: Find the round-off error in storing the number 752.6835 using a four digit mantissa.

$\Rightarrow x = 752.6835$

$x = 0.75268835 \times 10^{3} = (0.7526 + 0.0000835)10^{3}$ $\quad 0.835\times10^{-4}$

True $x = 0.7526 \times 10^{3} + 0.835 \times 10^{(3-4)}$

$= 0.7526 \times 10^{3} + 0.835 \times 10^{-1}$

for chopping:

Approximate, $x = 0.7526 \times 10^{3}$

Error $= gx10^{E-d} = 0.835 \times 10^{-1}$

for symmetric round-off:

Error $= (g-1)\ 10^{(E-d)}$

$= (0.835 - 1) \times 10^{-1}$

$= -0.165 \times 10^{-1}$

$= -0.0165.$

$\therefore$ Approximate, $x = 0.7527 \times 10^{3}$

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Roots of Nonlinear Equations.

3 types of equations:

1. Algebric equation

$y = f(x) \rightarrow f(x,y) = 0$

$3x + 4y = 0$

2. Polynomical equation

$ax^2 + bx + c = 0$

$x^3 - 4x + 4 = 0$

3. Transcendental equation

$\sin x, \cos x, e^x, \log x.$

Roots / zeros: The values of x which satisfy the equation $f(x) = 0$ are called roots of the equation / the zeros of $f(x)$.

Iterative method:

1. Bracketing methods.
   - Bisection method
   - [a, b]
   - False position / Regula-Falsi method.
2. Open end methods.

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Biseation method: The biseation method is one of the simplest and most reliable of iterative methods for the solution of nonlinear equations. This method also known as binary chopping or half-interval method.

Algorithm:

1. Choose two real numbers a and b such that $f(a) * f(b) < 0$.
2. Define root $c = \frac{a+b}{2}$
3. Find $f(c)$.
4. If $f(a) * f(c) \leq 0$ then
   set $b = c$
   else $a = c$

Return to step 1 until finding the root matched twice.

Example-1: $f(x) = 3x - \cos x - 1$

$a = 0$, $f(a) = f(0) = 3 \cdot 0 - \cos(0) - 1 = -2$

$b = 1$, $f(b) = f(1) = 3 \cdot 1 - \cos(1) - 1 = 1.46$

$\therefore f(a) * f(b) = -2 \times 1.46 = -2.92 < 0$

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step	a	b	f(a)	f(b)	root $c = \frac{a+b}{2}$	f(c)
1	0	1	-2	1.46	0.5	-0.38
2	0.5	1	-0.38	1.46	0.75	0.52
3	0.5	0.75	-0.38	0.52	0.625	0.06
4	0.5	0.625	-0.38	0.064	0.5625	-0.16
5	0.5625	0.625	-0.16	0.064	0.59	-0.06
6	0.59	0.625	-0.06	0.064	0.61	0.01
7	0.59	0.61	-0.06	0.01	0.6	-0.025
8	0.6	0.61	-0.025	0.01	0.615	0.01

$\therefore$ root $c = 0.61$

Example-2 : $f(x) = x^3 - 4x - 9$

step	a	b	f(a)	f(b)	$c = \frac{a+b}{2}$	f(c)
1	2	3	-9	6	2.5	-3.375
2	2.5	3	-3.375	6	2.75	0.797
3	2.5	2.75	-3.375	0.797	2.625	-1.412
4	2.625	2.75	-1.412	0.797	2.688	-0.33
5	2.688	2.75	-0.33	0.797	2.72	0.244
6	2.688	2.72	-0.33	0.244	2.70	-0.045
7	2.70	2.72	-0.045	0.244	2.71	0.0086

root = 2.70

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Example 3 - $f(x) = x^3 - 4$

$a = 1, f(a) = f(1) = 1^3 - 4 = -3$

$b = 2, f(b) = f(2) = 2^3 - 4 = 4$

$f(a) * f(b) = -3 \times 4 = -12 < 0$

step	a	b	f(a)	f(b)	$c = \frac{a+b}{2}$	f(c)
1	1	2	-3	4	1.5	-0.625
2	1.5	2	-0.625	4	1.75	1.359
3	1.5	1.75	-0.625	1.359	1.61	0.173
4	1.5	1.61	-0.625	0.173	1.56	-0.204
5	1.56	1.61	-0.204	0.173	1.59	0.0107
6	1.56	1.59	-0.204	0.0107	1.57	-0.056
7	1.58	1.59	-0.056	0.0107	1.59	0.0107

Root = 1.59

False - Position method / Regula - falsi method.

The false position method is a numerical technique for finding the approximate root of an equation by using a straight - line interpolation between two points having opposite signs.

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Algorithm:

1. Choose two real numbers a and b such that $f(a) * f(b) < 0$
2. Define root $c = \frac{a * f(b) - b * f(a)}{f(b) - f(a)}$
3. Find $f(c)$
4. If $f(a) * f(c) < 0$ then
   set $b = c$
   else $a = c$

   Return to step 1 until finding the root matched twice.

Example 1 - $f(x) = 3x - \cos x - 1$

$a = 0$; $f(0) = 0 - 0 - 1 = -1$

$b = 1$, $f(1) = 3 - \cos(1) - 1 = 1.46$

$f(a) * f(b) = (-1) \times 1.46 = -1.46 < 0$

Step	a	b	f(a)	f(b)	$c = \frac{a \times f(b) - b \times f(a)}{f(b) - f(a)}$	f(c)
1	0	1	-1	1.46	0.41	-0.667
2	0.41	1	-0.667	1.46	0.599	-0.029
3	0.599	1	-0.029	1.46	0.61	0.01
4	0.599	0.61	-0.029	0.01	0.61	0.01

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∴ root = 0.61

Example:2- $f(x) = x^3 - 4x - 9$

$a = 2,\ f(2) = 2^3 - 4\times2 - 9 = -9$

$b = 3,\ f(3) = 3^3 - 4\times3 - 9 = 6$

$f(a) \times f(b) = -54 < 0$

Step	a	b	f(a)	f(b)	$c = \frac{a*f(b) - b*f(a)}{f(b) - f(a)}$	f(c)
1	2	3	-9	6	2.6	-1.824
2	2.6	3	-1.824	6	2.69	-0.205
3	2.69	3	-0.205	6	2.70	-0.117
4	2.70	3	-0.117	6	2.70	-0.117

∴ root = 2.70

Example:3- $f(x) = x^3 - 4$

Step	a	b	f(a)	f(b)	$c = \frac{a*f(b) - b*f(a)}{f(b) - f(a)}$	f(c)
1	1	2	-3	4	1.43	-1.076
2	1.43	2	-1.076	4	1.55	-0.276
3	1.55	2	-0.276	4	1.58	-0.056
4	1.58	2	-0.056	4	1.58	-0.056

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∴ root = 1.56

Open method:

1. Fixed Point / one-Point Iterative method
2. Newton - Raphson method.
3. Modified Newton - Raphson method.
4. Secant method.

Fixed point:

The fixed point method is a numerical method used to find the solution of an equation by rewriting it in the form:

$x = g(x)$

Starting with an initial guess, the next approximation is calculate repeatedly using:

$x_{n+1} = g(x_n)$

until the values become nearly equal.

Algorithm:

Step-1: Given an equation in $f(x) = 0$

Step-2: Convert $f(x) = 0$ into the from of $x = g(x)$

Step-3: Let the initial guess ($\log x, \sin x, \cos x, e^x$ = 3.2 simple = 0.5)

Do $x_{i+1} = g(x_i)$

while (none of the convergence iteration is matched.)

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Example-1: $f(x) = 3x - \cos x - 1$

Step-1: $f(x) = 0$

$\Rightarrow 3x - \cos x - 1 = 0$

$\Rightarrow 3x = 1 + \cos x$

$\Rightarrow x = \frac{1 + \cos x}{3}$

Step-2:

$\therefore x = g(x)$

where $g(x) = \frac{1 + \cos x}{3}$

Step-3:

$x_0 = g(3.2)$

$= \frac{1 + \cos(3.2)}{3} = 0$

$x_1 = g(x_0)$

$= g(0)$

$= \frac{1 + \cos(0)}{3} = 0.67$

$x_2 = g(x_1)$

$= g(0.67)$

$= \frac{1 + \cos(0.67)}{3} = 0.59$

$x_3 = g(x_2)$

$= g(0.59)$

$= \frac{1 + \cos(0.59)}{3} = 0.60$

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$x_4 = g(x_3)$

$= g(0.60)$

$= \frac{1+\cos(0.60)}{3} = 0.61$

$\therefore$ Root $= 0.61$

Example-2: $f(x) = 2x^3 - 2x - 5$

Step-1: $f(x) = 0$

$\Rightarrow 2x^3 - 2x - 5 = 0$

$\Rightarrow 2x^3 = 2x + 5$

$\Rightarrow x^3 = \frac{2x+5}{2}$

$\Rightarrow x = \sqrt[3]{\frac{2x+5}{2}} = \left(\frac{2x+5}{2}\right)^{\frac{1}{3}}$

Step-2:

$\therefore x = g(x)$

where $g(x) = \left(\frac{2x+5}{2}\right)^{\frac{1}{3}}$

Step-3:

$x_0 = g(0.5)$

$= \left(\frac{2(0.5)+5}{2}\right)^{\frac{1}{3}} = 1.44$

$x_1 = g(x_0)$

$= g(1.44)$

$= \left(\frac{2 \times 1.44 + 5}{2}\right)^{\frac{1}{3}} = 1.58$

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$x_2 = g(x_1)$

$= g(1.58)$

$= \left(\frac{2 \times 1.58 + 5}{2}\right)^{\frac{1}{3}} = 1.59$

$x_3 = g(x_2)$

$= g(1.59)$

$= \left(\frac{2 \times 1.59 + 5}{2}\right)^{\frac{1}{3}} = 1.59$

$\therefore$ Root $= 1.59$

Example-3: $f(x) = x^2 - x - 1$

Step-1: $f(x) = 0$

$\Rightarrow x^2 - x - 1 = 0$

$\Rightarrow x^2 = x + 1$

$\Rightarrow x = \sqrt{x+1} = (x+1)^{\frac{1}{2}}$

Step-2: $x = g(x)$

where $g(x) = (x+1)^{\frac{1}{2}}$

Step-3:

$x_0 = g(0.5)$

$= (0.5+1)^{\frac{1}{2}} = 1.22$

$x_1 = g(x_0)$

$= g(1.22)$

$= (1.22+1)^{\frac{1}{2}} = 1.49$

$x_2 = g(x_1)$

$= g(1.49)$

$= (1.49+1)^{\frac{1}{2}} = 1.58$

$x_3 = g(x_2)$

$= g(1.58)$

$= (1.58+1)^{\frac{1}{2}} = 1.61$

$x_4 = g(x_3)$

$= g(1.61)$

$= (1.61+1)^{\frac{1}{2}} = 1.61$

$\therefore$ Root $= 1.61$

Example-4: $f(x) = x^3 - 4x - 9$

Step-1: $f(x) = 0$

$\Rightarrow x^3 - 4x - 9 = 0$

$\Rightarrow x = (4x+9)^{\frac{1}{3}}$

Step-2 $g(x) = (4x+9)^{\frac{1}{3}}$

Step-3: $x_0 = g(0.5)$

$= (4\times 0.5+9)^{\frac{1}{3}} = 2.22$

$x_1 = g(x_0)$

$= (4\times 2.22+9)^{\frac{1}{3}} = 2.61$

$x_2 = g(x_1)$

$= (4\times 2.61+9)^{\frac{1}{3}} = 2.69$

$x_3 = g(x_2) = (4\times 2.69+9)^{\frac{1}{3}} = 2.70$

$x_4 = (4\times 2.70+9)^{\frac{1}{3}}$

$= 2.70$

$\therefore$ Root $= 2.70$

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Newton - Raphson method :

The newton - raphson method is a numerical method used to find the approximate root of an equation by using tangent lines.

Starting from an initial guess x_0, the next approximation is calculated by:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Algorithm :

Step 1 : Find $f'(x_n)$

Step 2 : Find a, b. so that

$$f(a) * f(b) < 0$$

Step 3 : Assume $x_0 = a$

Step 4 : Find out $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

Find $x_1, x_2, x_3 \ldots\ldots x_n$ until any two sucessive values are equal.

Example 1 :

$$f(x) = 3x - \cos x - 1. \quad \text{——} (1)$$

$$f'(x) = 3 - (-\sin x) - 0$$

$$\therefore f'(x) = 3 + \sin x \quad \text{——} (2)$$

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$a = 0$, $f(a)$, $f(0) = 3 \times 0 - \cos(0) - 1 = -2$

$b = 1$, $f(b)$, $f(1) = 3 \times 1 - \cos(1) - 1 = 1.46$

$\therefore f(a) * f(b) = -2 * 1.46$

$= -2.92 < 0$

Now, Let $x_0 = a = 0$

step	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	0	-2	3	$x_1 = 0 - \frac{-2}{3} = 0.67$
1	0.67	0.23	3.62	$x_2 = 0.67 - \frac{0.23}{3.62} = 0.61$
2	0.61	0.01	3.57	$x_3 = 0.61 - \frac{0.01}{3.57} = 0.61$

$\therefore$ Root = 0.61

Example-2 :

$f(x) = x^3 - 4x + 9$

$f'(x) = 3x^2 - 4$

$a = 2$, $f(a) = f(2) = (2)^3 - 4 \times 2 + 9 = -9$

$b = 3$, $f(b) = f(3) = (3)^3 - 4 \times 3 + 9 = 6$

$\therefore f(a) * f(b) = -9 \times 6$

$= -54 < 0$

Now, Let, $x_0 = a = 2$

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step	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	2	-9	8	$x_1 = 2 - \frac{-9}{8} = 3.12$
1	3.12	8.89	25.20	$x_2 = 3.12 - \frac{8.89}{25.20} = 2.77$
2	2.77	1.17	19.02	$x_3 = 2.77 - \frac{1.17}{19.02} = 2.70$
3	2.70	-0.117	17.87	$x_4 = 2.70 - \frac{-0.117}{17.87} = 2.70$

$\therefore$ Root $= 2.70$

Example-3:

$f(x) = x^3 - 4$

$f'(x) = 3x^2$

$a = 1$, $f(a) = f(1) = 1 - 4 = -3$

$b = 2$, $f(b) = f(2) = 8 - 4 = 4$

$\therefore f(a) * f(b) = -3 \times 4 = -12 < 0$

Now, let $x_0 = a = 1$

$\therefore$ Root $= 1.59$

step	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	1	-3	3	$x_1 = 1 - \frac{-3}{3} = 2$
1	2	4	12	$x_2 = 2 - \frac{4}{12} = 1.67$
2	1.67	0.66	8.36	$x_3 = 1.67 - \frac{0.66}{8.36} = 1.59$
3	1.59	0.019	7.58	$x_4 = 1.59 - \frac{0.019}{7.58} = 1.59$

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